Department of Physics, FNSPE, CTU in Prague

02YELMA - Electricity and Magnetism

Midterm 1

27.3.2026 / 16:00 - 17:30

Q1: A solid insulating sphere of radius R carries a spherically symmetric, non-uniform charge distribution given by $\rho(r) = \rho_0 \left(1 - \frac{r}{R}\right)$, where $0 < r < R$, $\rho(r) = 0$ for $r \geq R$, and ρ_0 is a constant. Calculate the electric field $\vec{E}(r)$ as a function of the distance r from the center of the sphere, for $0 < r < R$ and $r \geq R$. — *1 pt*

Q2: Two large, finite, parallel conducting plates of area A are placed a distance d apart in vacuum. One plate carries a total charge $+Q$, and the other carries $-Q$. The plates are electrically isolated and no the field is considered to be always uniform between the plates. The electric field outside the region between the plates is negligible.

(a) Find the electrostatic energy stored in the system by using the integral $U = \frac{\epsilon_0}{2} \int\limits_{\text{all space}} E^2 d\tau$, where ϵ_0 is a constant, $\vec{E}$ is the electric field, and $d\tau$ is the volume element. — *0.6 pts*

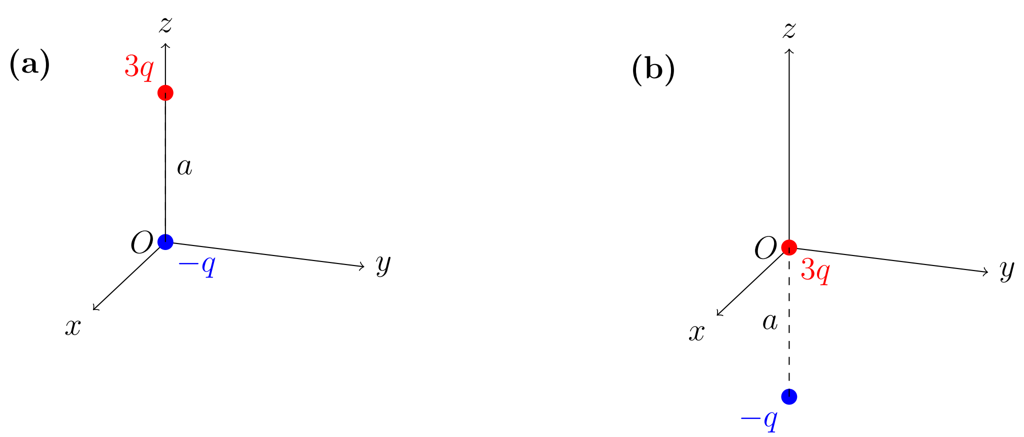
(b) Suppose the distance between the plates is increased by a small amount Δx with the charges remaining constant. How much work must be done in this process? — *0.4 pts*

Q3: A thin rod of length L lies along the x-axis extending from $x = 0$ to $x = L$. The rod carries a nonuniform linear charge density given by $\lambda(x) = \lambda_0 \frac{x}{L}$ where λ_0 is a constant with units of charge per unit length. What is the net electric potential V at $x = -d$ due to the entire rod? Take the reference $V(\infty) = 0$. — *1 pt*
Given: $\int_0^{x_0} \frac{x\,dx}{a+x} = x_0 - a \ln\left(\frac{a+x_0}{a}\right)$, where $x_0 > 0$ and $a > 0$ are constants.

Q4: The first two terms of multipole expansion of the potential is given by

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0}\left(\frac{Q}{r} + \frac{\vec{p}\cdot\hat{r}}{r^2} + \dots\right)$$

where Q is the monopole moment, $\vec{p}$ is the dipole moment. The spherical coordinates are given by (r, θ, ϕ), where r is the distance from the origin, θ is the polar angle, and ϕ is the azimuthal angle. Two charges are placed in two different arrangements as shown below:



For each arrangement, determine the monopole moment *(0.3 pts)*, the dipole moment *(0.3 pts)*, and the approximate potential in spherical coordinates up to the dipole term *(0.4 pts)*.